

Лабораторная работа № 2

Измерение и тестирование пропускной способности сети. Интерактивный эксперимент

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Основной целью работы является знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

1. Установить на виртуальную машину mininet iPerf3 и дополнительное программное обеспечения для визуализации и обработки данных.
2. Провести ряд интерактивных экспериментов по измерению пропускной способности с помощью iPerf3 с построением графиков.

Выполнение лабораторной работы

Установка необходимого программного обеспечения

```
mininet@mininet-vm:~$ sudo dhclient eth1
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
    ether 08:00:27:d9:0b:a8 txqueuelen 1000 (Ethernet)
    RX packets 65 bytes 10330 (10.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 49 bytes 8989 (8.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    ether 08:00:27:03:cc:2d txqueuelen 1000 (Ethernet)
    RX packets 1 bytes 590 (590.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1 bytes 342 (342.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 120 bytes 9192 (9.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 120 bytes 9192 (9.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Figure 1: Подключение к mininet по ssh

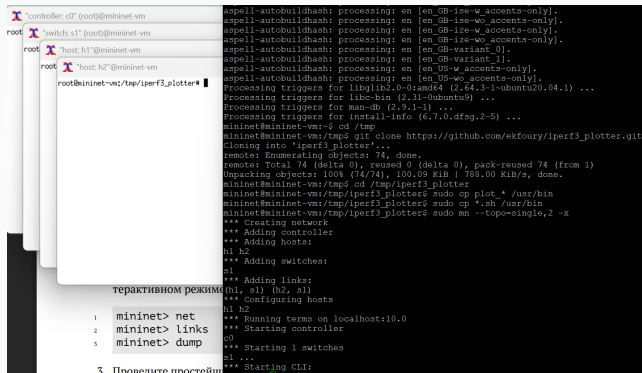
Установка необходимого программного обеспечения

```
Get:27 http://us.archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [548 kB]
Get:28 http://us.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [604 B]
Get:29 http://us.archive.ubuntu.com/ubuntu focal-updates/universe i386 Packages [823 kB]
Get:30 http://us.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1,262 kB]
Get:31 http://us.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [303 kB]
Get:32 http://us.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [29.3 kB]
Get:33 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [29.7 kB]
Get:34 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse i386 Packages [8,440 B]
Get:35 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [8,316 B]
Get:36 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [688 B]
Get:37 http://us.archive.ubuntu.com/ubuntu focal-backports/main i386 Packages [36.1 kB]
Get:38 http://us.archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [45.7 kB]
Get:39 http://us.archive.ubuntu.com/ubuntu focal-backports/main Translation-en [16.3 kB]
Get:40 http://us.archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [1,420 B]
Get:41 http://us.archive.ubuntu.com/ubuntu focal-backports/universe i386 Packages [13.8 kB]
Get:42 http://us.archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [25.0 kB]
Get:43 http://us.archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [16.3 kB]
Get:44 http://us.archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B]
Fetched 24.5 MB in 6s (4,120 kB/s)
Reading package lists... Done
```

Установка необходимого программного обеспечения

```
The following NEW packages will be installed:
  iperf3 libiperf0 libsctp1
0 upgraded, 3 newly installed, 0 to remove and 393 not upgraded.
Need to get 94.1 kB of archives.
After this operation, 331 kB of additional disk space will be used.
Do you want to continue? [Y/n] yes
Get:1 http://us.archive.ubuntu.com/ubuntu focal/main amd64 libsctp1 amd64 1.0.18+
dfsg-1 [7,876 B]
Get:2 http://us.archive.ubuntu.com/ubuntu focal/universe amd64 libiperf0 amd64 3.
7-3 [72.0 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu focal/universe amd64 iperf3 amd64 3.7-3
[14.2 kB]
Fetched 94.1 kB in 1s (107 kB/s)
Selecting previously unselected package libsctp1:amd64.
(Reading database ... 101729 files and directories currently installed.)
Preparing to unpack .../libsctp1_1.0.18+dfsg-1_amd64.deb ...
Unpacking libsctp1:amd64 (1.0.18+dfsg-1) ...
Selecting previously unselected package libiperf0:amd64.
Preparing to unpack .../libiperf0_3.7-3_amd64.deb ...
Unpacking libiperf0:amd64 (3.7-3) ...
Selecting previously unselected package iperf3.
Preparing to unpack .../iperf3_3.7-3_amd64.deb ...
Unpacking iperf3 (3.7-3) ...
Setting up libsctp1:amd64 (1.0.18+dfsg-1) ...
Setting up libiperf0:amd64 (3.7-3) ...
Setting up iperf3 (3.7-3) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9) ...
```

Figure 3: Развертывание iperf3_plotter



```
controller: c0 (root)@mininet-vm
root@mininet-vm:~# switch: s1 (root)@mininet-vm
root@mininet-vm:~# host: h1 (root)@mininet-vm
root@mininet-vm:~# host: h2 (root)@mininet-vm
root@mininet-vm:~# root@mininet-vm:~/tmp/iperf3_plotter#

asperf3-autobuildhash: processing: en [en_GB-ise-w_accents-only].
asperf3-autobuildhash: processing: en [en_GB-ise-wo_accents-only].
asperf3-autobuildhash: processing: en [en_GB-ize-w_accents-only].
asperf3-autobuildhash: processing: en [en_GB-ize-wo_accents-only].
asperf3-autobuildhash: processing: en [en_GB-variant_0].
asperf3-autobuildhash: processing: en [en_GB-variant_1].
asperf3-autobuildhash: processing: en [en_US-w_accents-only].
asperf3-autobuildhash: processing: en [en_US-wo_accents-only].
Processing triggers for libc-bin (2.31-0ubuntu9) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
mininet@mininet-vm:~$ cd /tmp
mininet@mininet-vm:~/tmp$ git clone https://github.com/ekfoury/iperf3_plotter.git
Cloning into 'iperf3_plotter'...
remote: Enumerating objects: 74, done.
remote: Total 74 (delta 0), reused 0 (delta 0), pack-reused 74 (from 1)
Unpacking objects: 100% (74/74), 100.09 KiB | 788.00 KiB/s, done.
mininet@mininet-vm:~/tmp$ cd /tmp/iperf3_plotter
mininet@mininet-vm:~/tmp/iperf3_plotter$ sudo cp plot.* /usr/bin
mininet@mininet-vm:~/tmp/iperf3_plotter$ sudo cp *.sh /usr/bin
mininet@mininet-vm:~/tmp/iperf3_plotter$ sudo mn --topo=single,2 -x
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Running terms on localhost:10.0
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
```

терактивном режиме

```
1 mininet> net
2 mininet> links
3 mininet> dump
```

3 Проведите простейший

Figure 4: Задание простейшей топологии

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=4087>
<Host h2: h2-eth0:10.0.0.2 pid=4091>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=4096>
<Controller c0: 127.0.0.1:6653 pid=4080>
mininet> █
```

Figure 5: Параметры запущенной в интерактивном режиме топологии

```
host: h2" @mininet-vm
-----
Server listening on 5201
Accepted connection from 10.0.0.1, port 35156
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 35164
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-1.00 sec  6.05 GBytes  51.9 Gbits/sec
[ 7] 1.00-2.00 sec  6.07 GBytes  52.1 Gbits/sec
[ 7] 2.00-3.00 sec  5.67 GBytes  49.7 Gbits/sec
[ 7] 3.00-4.00 sec  4.83 GBytes  41.5 Gbits/sec
[ 7] 4.00-5.00 sec  5.71 GBytes  49.0 Gbits/sec
[ 7] 5.00-6.00 sec  5.15 GBytes  44.2 Gbits/sec
[ 7] 6.00-7.00 sec  5.61 GBytes  48.2 Gbits/sec
[ 7] 7.00-8.00 sec  5.59 GBytes  48.0 Gbits/sec
[ 7] 8.00-9.00 sec  5.42 GBytes  46.6 Gbits/sec
[ 7] 9.00-10.00 sec 5.96 GBytes  51.2 Gbits/sec
[ 7] 10.00-10.00 sec 2.00 MBytes  42.6 Gbits/sec
-----
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.00 sec 56.1 GBytes  48.2 Gbits/sec
-----
Server listening on 5201

host: h1" @mininet-vm
-----
oot@mininet-vm:/tmp/iperf3_plotter# iperf3 -c 10.0.0.2
connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 35164 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec  6.05 GBytes  51.9 Gbits/sec  14    20.1 MBytes
[ 7] 1.00-2.00 sec  6.07 GBytes  52.1 Gbits/sec   0    20.1 MBytes
[ 7] 2.00-3.00 sec  5.67 GBytes  49.7 Gbits/sec   0    20.1 MBytes
[ 7] 3.00-4.00 sec  4.83 GBytes  41.5 Gbits/sec   0    20.1 MBytes
[ 7] 4.00-5.00 sec  5.71 GBytes  49.0 Gbits/sec  47    14.1 MBytes
[ 7] 5.00-6.00 sec  5.15 GBytes  44.2 Gbits/sec   0    14.1 MBytes
[ 7] 6.00-7.00 sec  5.61 GBytes  48.2 Gbits/sec   0    14.1 MBytes
[ 7] 7.00-8.00 sec  5.59 GBytes  48.0 Gbits/sec   0    14.1 MBytes
[ 7] 8.00-9.00 sec  5.42 GBytes  46.6 Gbits/sec   0    14.1 MBytes
[ 7] 9.00-10.00 sec 5.97 GBytes  51.2 Gbits/sec  47    9.86 MBytes
-----
[ ID] Interval      Transfer      Bitrate      Retr
[ 7] 0.00-10.00 sec 56.1 GBytes  48.2 Gbits/sec  108
[ 7] 0.00-10.00 sec 56.1 GBytes  48.2 Gbits/sec
-----
perf Done.
oot@mininet-vm:/tmp/iperf3_plotter#
```

Figure 6: Тестовое соединение между хостами

```
mininet@mininet-vm: /tmp/iperf3_plotter
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=1382>
<Controller c0: 127.0.0.1:6653 pid=1366>
mininet> h2 iperf3 -s &
mininet> h1 iperf3 -c h2
Connecting to host 10.0.0.2, port 5201
[ 5] local 10.0.0.1 port 43470 connected to 10.0.0.2 port 5201
[ ID] Interval            Transfer          Bitrate          Retr  Cwnd
[ 5]  0.00-1.00    sec   6.14 GBytes     52.7 Gbits/sec      0   1.39 MBytes
[ 5]  1.00-2.00    sec   6.08 GBytes     52.2 Gbits/sec      0   3.64 MBytes
[ 5]  2.00-3.00    sec   6.17 GBytes     53.0 Gbits/sec      0   5.25 MBytes
[ 5]  3.00-4.00    sec   6.08 GBytes     52.2 Gbits/sec      0   5.79 MBytes
[ 5]  4.00-5.00    sec   5.94 GBytes     51.0 Gbits/sec      0   6.08 MBytes
[ 5]  5.00-6.00    sec   6.17 GBytes     53.0 Gbits/sec      0   6.70 MBytes
[ 5]  6.00-7.00    sec   6.03 GBytes     51.8 Gbits/sec      0   8.15 MBytes
[ 5]  7.00-8.00    sec   6.10 GBytes     52.4 Gbits/sec      0   8.15 MBytes
[ 5]  8.00-9.00    sec   6.09 GBytes     52.3 Gbits/sec      0   8.15 MBytes
[ 5]  9.00-10.00   sec   6.03 GBytes     51.8 Gbits/sec    137   5.74 MBytes
-----
[ ID] Interval            Transfer          Bitrate          Retr
[ 5]  0.00-10.00   sec   60.8 GBytes     52.2 Gbits/sec    137
[ 5]  0.00-10.00   sec   60.8 GBytes     52.2 Gbits/sec
                                     sender
                                     receiver

iperf Done.
mininet> █
```

Figure 7: Эксперимент в интерфейсе mininet

```
mininet> h2 killall iperf3
warning: this system does not seem to support IPv6 - trying IPv4
iperf3: error - unable to start listener for connections: Address already in use
iperf3: exiting
Accepted connection from 10.0.0.1, port 54504
[ 5] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 54506
[ ID] Interval           Transfer     Bitrate
[ 5]  0.00-1.00    sec   5.76 GBytes  49.5 Gbits/sec
[ 5]  1.00-2.00    sec   5.72 GBytes  49.1 Gbits/sec
[ 5]  2.00-3.00    sec   5.72 GBytes  49.2 Gbits/sec
[ 5]  3.00-4.00    sec   5.80 GBytes  49.8 Gbits/sec
[ 5]  4.00-5.00    sec   5.96 GBytes  51.2 Gbits/sec
[ 5]  5.00-6.00    sec   5.83 GBytes  50.1 Gbits/sec
[ 5]  6.00-7.00    sec   5.92 GBytes  50.9 Gbits/sec
[ 5]  7.00-8.00    sec   5.70 GBytes  49.0 Gbits/sec
[ 5]  8.00-9.00    sec   5.56 GBytes  47.8 Gbits/sec
[ 5]  9.00-10.00   sec   5.76 GBytes  49.5 Gbits/sec
[ 5] 10.00-10.00   sec     640 KBytes  52.4 Gbits/sec
-----
[ ID] Interval           Transfer     Bitrate
[ 5]  0.00-10.00   sec   57.7 GBytes  49.6 Gbits/sec
-----
Server listening on 5201
-----
iperf3: interrupt - the server has terminated
mininet>
```

Figure 8: Завершение процесса на сервере

```
"host: h2"@mininet-vm
Server listening on 5201

iperf3: interrupt - the server has terminated
root@mininet-vm:/tmp/iperf3_plotter# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4

Server listening on 5201

Accepted connection from 10.0.0.1, port 50950
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 50954
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-1.00    sec 6.12 GBytes 52.6 Gbits/sec
[ 7] 1.00-2.00    sec 6.12 GBytes 52.6 Gbits/sec
[ 7] 2.00-3.00    sec 5.39 GBytes 46.3 Gbits/sec
[ 7] 3.00-4.00    sec 6.09 GBytes 52.3 Gbits/sec
[ 7] 4.00-5.00    sec 6.11 GBytes 52.5 Gbits/sec
[ 7] 5.00-5.00    sec 768 KBytes 43.1 Gbits/sec

[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-5.00    sec 29.8 GBytes 51.2 Gbits/sec                                     receiver

Server listening on 5201

[
[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-10.00   sec 56.1 GBytes 48.2 Gbits/sec 108
[ 7] 0.00-10.00   sec 56.1 GBytes 48.2 Gbits/sec
sender
receiver

iperf Done.
root@mininet-vm:/tmp/iperf3_plotter# iperf3 -c 10.0.0.2 -t 5
connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 50954 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 7] 0.00-1.00    sec 6.12 GBytes 52.6 Gbits/sec 10      443 KBytes
[ 7] 1.00-2.00    sec 6.12 GBytes 52.5 Gbits/sec 0       532 KBytes
[ 7] 2.00-3.00    sec 5.39 GBytes 46.3 Gbits/sec 0       677 KBytes
[ 7] 3.00-4.00    sec 6.09 GBytes 52.3 Gbits/sec 0       757 KBytes
[ 7] 4.00-5.00    sec 6.11 GBytes 52.5 Gbits/sec 92      652 KBytes

[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-5.00    sec 29.8 GBytes 51.2 Gbits/sec 102
[ 7] 0.00-5.00    sec 29.8 GBytes 51.2 Gbits/sec
sender
receiver

iperf Done.
root@mininet-vm:/tmp/iperf3_plotter#
```

Figure 9: Указание периода времени передачи

```
-----
^Ciperf3: interrupt - the server has terminated
root@mininet-virtual-machine:~# iperf3 -s -i 2
warning: this system does not seem to support IPv6 - trying IPv4
-----
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 43772
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 43772
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-2.00 sec  12.5 GBytes 53.7 Gbits/sec
[ 7] 2.00-4.00 sec  11.4 GBytes 49.1 Gbits/sec
[ 7] 4.00-6.00 sec  12.1 GBytes 51.9 Gbits/sec
[ 7] 6.00-8.00 sec  11.8 GBytes 50.7 Gbits/sec
[ 7] 8.00-10.00 sec 12.2 GBytes 52.2 Gbits/sec
[ 7] 10.00-10.00 sec 512 KBytes 24.1 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.00 sec 60.0 GBytes 51.5 Gbits/sec
-----
Server listening on 5201
-----
[ 7] 3.00-4.00 sec 6.09 GBytes 52.3 Gbits/sec 0 757 KBytes
[ 7] 4.00-5.00 sec 6.11 GBytes 52.5 Gbits/sec 92 652 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-5.00 sec 29.8 GBytes 51.2 Gbits/sec 102
[ 7] 0.00-5.00 sec 29.8 GBytes 51.2 Gbits/sec
-----
iperf Done.
root@mininet-virtual-machine:~# iperf3 -c 10.0.0.2 -i 2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 43778 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 7] 0.00-2.00 sec  12.5 GBytes 53.7 Gbits/sec 17 5.70 MBytes
[ 7] 2.00-4.00 sec  11.4 GBytes 49.1 Gbits/sec 2 5.70 MBytes
[ 7] 4.00-6.00 sec  12.1 GBytes 51.9 Gbits/sec 0 5.71 MBytes
[ 7] 6.00-8.00 sec  11.8 GBytes 50.7 Gbits/sec 0 5.72 MBytes
[ 7] 8.00-10.00 sec 12.2 GBytes 52.2 Gbits/sec 0 5.73 MBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-10.00 sec 60.0 GBytes 51.5 Gbits/sec 19
[ 7] 0.00-10.00 sec 60.0 GBytes 51.5 Gbits/sec
-----
iperf Done.
```

Figure 10: Выполнения теста пропускной способности с 2-секундным интервалом

```
-----  
Server listening on 5201  
-----  
^Ciperf3: interrupt - the server has terminated  
root@mininet-vsp/tmp/iperf3_plotter# iperf3 -s  
warning: this system does not seem to support IPv6 - trying IPv4  
-----  
Server listening on 5201  
-----  
Accepted connection from 10.0.0.1, port 39356  
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 39360  
[ ID] Interval      Transfer      Bitrate  
[ 7] 0.00-1.00 sec  5.94 GBytes  51.0 Gbits/sec  
[ 7] 1.00-2.00 sec  6.07 GBytes  52.1 Gbits/sec  
[ 7] 2.00-2.67 sec  3.99 GBytes  51.2 Gbits/sec  
-----  
[ ID] Interval      Transfer      Bitrate  
[ 7] 0.00-2.67 sec  16.0 GBytes  51.5 Gbits/sec  
-----  
Server listening on 5201  
-----  
X "host: h1" @ mininet-vm  
-----  
[ 7] 2.00-4.00 sec  11.4 GBytes  49.1 Gbits/sec  2  5.70 MBytes  
[ 7] 4.00-6.00 sec  12.1 GBytes  51.9 Gbits/sec  0  5.71 MBytes  
[ 7] 6.00-8.00 sec  11.8 GBytes  50.7 Gbits/sec  0  5.72 MBytes  
[ 7] 8.00-10.00 sec  12.2 GBytes  52.2 Gbits/sec  0  5.73 MBytes  
-----  
[ ID] Interval      Transfer      Bitrate      Retr  
[ 7] 0.00-10.00 sec  60.0 GBytes  51.5 Gbits/sec  19  
[ 7] 0.00-10.00 sec  60.0 GBytes  51.5 Gbits/sec  
-----  
iperf Done.  
root@mininet-vm:/tmp/iperf3_plotter# iperf3 -c 10.0.0.2 -n 16G  
connecting to host 10.0.0.2, port 5201  
[ 7] local 10.0.0.1 port 39360 connected to 10.0.0.2 port 5201  
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd  
[ 7] 0.00-1.00 sec  5.94 GBytes  51.0 Gbits/sec  13  5.70 MBytes  
[ 7] 1.00-2.00 sec  6.07 GBytes  52.1 Gbits/sec  0  5.70 MBytes  
[ 7] 2.00-2.67 sec  3.99 GBytes  51.2 Gbits/sec  0  5.70 MBytes  
-----  
[ ID] Interval      Transfer      Bitrate      Retr  
[ 7] 0.00-2.67 sec  16.0 GBytes  51.5 Gbits/sec  13  
[ 7] 0.00-2.67 sec  16.0 GBytes  51.5 Gbits/sec  
-----  
iperf Done.
```

Figure 11: Задание в тесте определённого объёма данных

```
Accepted connection from 10.0.0.1, port 59048
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 51628
[ ID] Interval      Transfer    Bitrate      Jitter    Lost/Total Datag
rans
[ 7] 0.00-1.00 sec  129 KBytes  1.05 Mbits/sec  0.045 ms  0/91 (0%)
[ 7] 1.00-2.00 sec  129 KBytes  1.05 Mbits/sec  0.050 ms  0/91 (0%)
[ 7] 2.00-3.00 sec  127 KBytes  1.04 Mbits/sec  0.020 ms  0/90 (0%)
[ 7] 3.00-4.00 sec  127 KBytes  1.04 Mbits/sec  0.026 ms  0/90 (0%)
[ 7] 4.00-5.04 sec  127 KBytes  1.01 Mbits/sec  0.015 ms  0/90 (0%)
[ 7] 5.04-6.00 sec  127 KBytes  1.08 Mbits/sec  0.027 ms  0/90 (0%)
[ 7] 6.00-7.02 sec  127 KBytes  1.02 Mbits/sec  0.034 ms  0/90 (0%)
[ 7] 7.02-8.00 sec  132 KBytes  1.10 Mbits/sec  0.020 ms  0/93 (0%)
[ 7] 8.00-9.05 sec  126 KBytes  984 Kbits/sec  0.016 ms  0/89 (0%)
[ 7] 9.05-10.00 sec 127 KBytes  1.10 Mbits/sec  0.017 ms  0/90 (0%)
-----
[ ID] Interval      Transfer    Bitrate      Jitter    Lost/Total Datag
rans
[ 7] 0.00-10.00 sec 1.25 MBytes  1.05 Mbits/sec  0.017 ms  0/904 (0%) rece
iver
-----
Server listening on 5201

X "host: h1" @mininet-vn
[ 7] local 10.0.0.1 port 51628 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Total Datagrams
[ 7] 0.00-1.00 sec  129 KBytes  1.05 Mbits/sec  91
[ 7] 1.00-2.00 sec  127 KBytes  1.04 Mbits/sec  90
[ 7] 2.00-3.00 sec  129 KBytes  1.06 Mbits/sec  91
[ 7] 3.00-4.00 sec  127 KBytes  1.04 Mbits/sec  90
[ 7] 4.00-5.04 sec  127 KBytes  1.01 Mbits/sec  90
[ 7] 5.04-6.00 sec  127 KBytes  1.08 Mbits/sec  90
[ 7] 6.00-7.02 sec  127 KBytes  1.02 Mbits/sec  90
[ 7] 7.02-8.00 sec  132 KBytes  1.10 Mbits/sec  93
[ 7] 8.00-9.05 sec  126 KBytes  984 Kbits/sec  89
[ 7] 9.05-10.00 sec 127 KBytes  1.10 Mbits/sec  90
-----
[ ID] Interval      Transfer    Bitrate      Jitter    Lost/Total Datag
rans
[ 7] 0.00-10.00 sec 1.25 MBytes  1.05 Mbits/sec  0.000 ms  0/904 (0%) send
er
[ 7] 0.00-10.00 sec 1.25 MBytes  1.05 Mbits/sec  0.017 ms  0/904 (0%) rece
iver

iperf Done.
root@mininet-vn:/tmp/iperf3_plotter#
```

Figure 12: Изменение протокола передачи данных

```
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-1.00 sec 5.75 GBytes 49.4 Gbits/sec
[ 7] 1.00-2.00 sec 5.73 GBytes 49.2 Gbits/sec
[ 7] 2.00-3.00 sec 5.48 GBytes 47.1 Gbits/sec
[ 7] 3.00-4.00 sec 5.59 GBytes 48.0 Gbits/sec
[ 7] 4.00-5.00 sec 5.67 GBytes 48.7 Gbits/sec
[ 7] 5.00-6.00 sec 5.59 GBytes 48.0 Gbits/sec
[ 7] 6.00-7.00 sec 5.69 GBytes 48.9 Gbits/sec
[ 7] 7.00-8.00 sec 5.74 GBytes 49.3 Gbits/sec
[ 7] 8.00-9.00 sec 5.63 GBytes 48.4 Gbits/sec
[ 7] 9.00-10.00 sec 5.10 GBytes 43.8 Gbits/sec
[ 7] 10.00-10.00 sec 1.00 MBytes 15.8 Gbits/sec
-----
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.00 sec 56.0 GBytes 48.1 Gbits/sec
-----
Server listening on 3250

X "host: h1" @mininet-vm
perf Done.
oot@mininet-vm:/tmp/iperf3_plotter# iperf3 -c 10.0.0.2 -p 3250
connecting to host 10.0.0.2, port 3250
[ 7] local 10.0.0.1 port 59140 connected to 10.0.0.2 port 3250
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec 5.75 GBytes 49.4 Gbits/sec 47 1.74 MBytes
[ 7] 1.00-2.00 sec 5.73 GBytes 49.2 Gbits/sec 0 1.81 MBytes
[ 7] 2.00-3.00 sec 5.48 GBytes 47.1 Gbits/sec 0 1.96 MBytes
[ 7] 3.00-4.00 sec 5.59 GBytes 48.0 Gbits/sec 0 2.02 MBytes
[ 7] 4.00-5.00 sec 5.67 GBytes 48.7 Gbits/sec 137 1.50 MBytes
[ 7] 5.00-6.00 sec 5.59 GBytes 48.0 Gbits/sec 0 1.55 MBytes
[ 7] 6.00-7.00 sec 5.69 GBytes 48.9 Gbits/sec 0 1.60 MBytes
[ 7] 7.00-8.00 sec 5.74 GBytes 49.3 Gbits/sec 0 1.65 MBytes
[ 7] 8.00-9.00 sec 5.63 GBytes 48.4 Gbits/sec 0 1.79 MBytes
[ 7] 9.00-10.00 sec 5.10 GBytes 43.8 Gbits/sec 0 1.79 MBytes
-----
[ ID] Interval      Transfer      Bitrate      Retr
[ 7] 0.00-10.00 sec 56.0 GBytes 48.1 Gbits/sec 184
[ 7] 0.00-10.00 sec 56.0 GBytes 48.1 Gbits/sec
-----
perf Done.
oot@mininet-vm:/tmp/iperf3_plotter#
```

Figure 13: Изменение номера порта для отправки/получения пакетов

```
Accepted connection from 10.0.0.1, port 53760
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 53776
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-1.00 sec  5.61 GBytes  48.2 Gbits/sec
[ 7] 1.00-2.00 sec  5.65 GBytes  48.5 Gbits/sec
[ 7] 2.00-3.00 sec  5.74 GBytes  49.3 Gbits/sec
[ 7] 3.00-4.00 sec  5.76 GBytes  49.5 Gbits/sec
[ 7] 4.00-5.00 sec  5.81 GBytes  49.9 Gbits/sec
[ 7] 5.00-6.00 sec  5.01 GBytes  43.0 Gbits/sec
[ 7] 6.00-7.00 sec  3.95 GBytes  33.9 Gbits/sec
[ 7] 7.00-8.00 sec  4.17 GBytes  35.8 Gbits/sec
[ 7] 8.00-9.00 sec  3.86 GBytes  33.2 Gbits/sec
[ 7] 9.00-10.00 sec 4.06 GBytes  34.9 Gbits/sec
[ 7] 10.00-10.00 sec 128 KBytes  5.76 Gbits/sec
- - - - -
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.00 sec 49.6 GBytes  42.6 Gbits/sec
root@mininet-vz:/tmp/iperf3_plotter# []

perf Done.
oot@mininet-vz:/tmp/iperf3_plotter# iperf3 -c 10.0.0.2
connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 53776 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec  5.61 GBytes  48.2 Gbits/sec  64    1.00 MBytes
[ 7] 1.00-2.00 sec  5.65 GBytes  48.5 Gbits/sec  0    1.00 MBytes
[ 7] 2.00-3.00 sec  5.74 GBytes  49.3 Gbits/sec  0    1.00 MBytes
[ 7] 3.00-4.00 sec  5.76 GBytes  49.5 Gbits/sec  0    1.00 MBytes
[ 7] 4.00-5.00 sec  5.81 GBytes  49.9 Gbits/sec  0    1.00 MBytes
[ 7] 5.00-6.00 sec  5.01 GBytes  43.0 Gbits/sec  0    1.00 MBytes
[ 7] 6.00-7.00 sec  3.95 GBytes  33.9 Gbits/sec  0    1.00 MBytes
[ 7] 7.00-8.00 sec  4.17 GBytes  35.8 Gbits/sec  0    1.00 MBytes
[ 7] 8.00-9.00 sec  3.86 GBytes  33.2 Gbits/sec  0    1.00 MBytes
[ 7] 9.00-10.00 sec 4.06 GBytes  34.9 Gbits/sec  0    1.00 MBytes
- - - - -
[ ID] Interval      Transfer      Bitrate      Retr
[ 7] 0.00-10.00 sec 49.6 GBytes  42.6 Gbits/sec  64
[ 7] 0.00-10.00 sec 49.6 GBytes  42.6 Gbits/sec
sender
receiver

perf Done.
oot@mininet-vz:/tmp/iperf3_plotter# []
```

Figure 14: Параметр обработки данных только от одного клиента с остановкой сервера по завершении теста

```
=====
Server listening on 5201
=====
Accepted connection from 10.0.0.1, port 40200
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 40200
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-1.00 sec  5.32 GBytes  50.8 Gbits/sec
[ 7] 1.00-2.01 sec  5.96 GBytes  50.8 Gbits/sec
[ 7] 2.01-3.00 sec  5.77 GBytes  49.9 Gbits/sec
[ 7] 3.00-4.00 sec  5.84 GBytes  50.2 Gbits/sec
[ 7] 4.00-5.00 sec  5.67 GBytes  48.7 Gbits/sec
[ 7] 5.00-6.00 sec  5.76 GBytes  49.5 Gbits/sec
[ 7] 6.00-7.00 sec  5.55 GBytes  47.7 Gbits/sec
[ 7] 7.00-8.00 sec  5.77 GBytes  49.6 Gbits/sec
[ 7] 8.00-9.00 sec  5.19 GBytes  44.6 Gbits/sec
[ 7] 9.00-10.00 sec  4.52 GBytes  38.8 Gbits/sec
[ 7] 10.00-10.00 sec  1.12 MBytes  17.6 Gbits/sec
=====
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.00 sec  55.9 GBytes  48.1 Gbits/sec
=====
receiver
=====
"host: h1" @mininet-vm
{
  "sender": true
},
"sum_received": {
  "start": 0,
  "end": 10.000552,
  "seconds": 10.000552,
  "bytes": 6007372864,
  "bits_per_second": 48056330221.771751,
  "sender": true
},
"cpu_utilization_percent": {
  "host_total": 99.569260570811466,
  "host_user": 1.3317227447714171,
  "host_system": 98.177537826040062,
  "remote_total": 32.983579094457063,
  "remote_user": 1.5105596494124146,
  "remote_system": 31.473023445044646
},
"sender_tcp_congestion": "cubic",
"receiver_tcp_congestion": "cubic"
}
=====
```

Figure 15: Экспорт результатов теста измерения пропускной способности iPerf3 в файл JSON

```
root@mininet-vm:/tmp/iperf3_plotter#  
root@mininet-vm:/tmp/iperf3_plotter# iperf3 -c 10.0.0.2 -J >/home/mininet/work/  
lab_iperf3/iperf_results.json  
root@mininet-vm:/tmp/iperf3_plotter# cd /home/mininet/work/lab_iperf3  
root@mininet-vm:/home/mininet/work/lab_iperf3# ls -l  
total 8  
-rw-r--r-- 1 root root 7777 Sep 26 05:06 iperf_results.json  
root@mininet-vm:/home/mininet/work/lab_iperf3#
```

Figure 16: Проверка создания файла iperf_results.json

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  e6d35fc8740a9b7b75a24b8bf3c07276
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth add mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  e6d35fc
8740a9b7b75a24b8bf3c07276
root@mininet-vm:~# logout
mininet@mininet-vm:~$
```

Figure 17: Исправление прав запуска X-соединения

```
> "retransmits": 0
>
> )
>
> }
> EOF
mininet@mininet-vm:~/work/lab_iperf3/results$ cd
mininet@mininet-vm:~$ ls -la
total 88
drwxr-xr-x 13 mininet mininet 4096 Sep 26 05:27 .
drwxr-xr-x  3 root    root    4096 Feb 10  2021 ..
-rw-r----- 1 mininet mininet 3831 Sep 26 05:14 .bash_history
-rw-r--r--  1 mininet mininet  220 Feb 10  2021 .bash_logout
-rw-r--r--  1 mininet mininet 3821 Sep 25 14:36 .bashrc
drwx----- 3 mininet mininet 4096 Sep 26 03:51 .cache
-rw-rw-r--  1 mininet mininet   85 Feb 10  2021 .gitconfig
-rw-rw-r--  1 mininet mininet 1640 Sep 26 05:26 iperf3_results.json
-rw-rw-r--  1 mininet mininet   1 Sep 26 05:27 iperf.csv
drwxrwxr-x 11 mininet mininet 4096 Sep 25 14:06 mininet
drwxrwxr-x 11 mininet mininet 4096 Feb 10  2021 mininet.orig
drwxrwxr-x 14 mininet mininet 4096 Feb 10  2021 oflops
drwxrwxr-x 11 mininet mininet 4096 Feb 10  2021 oftest
drwxrwxr-x 19 mininet mininet 4096 Feb 10  2021 openflow
drwxrwxr-x  7 mininet mininet 4096 Feb 10  2021 pox
-rw-r--r--  1 mininet mininet  807 Feb 10  2021 .profile
drwxrwxr-x  2 mininet mininet 4096 Sep 26 05:27 results
drwx----- 2 mininet mininet 4096 Sep 25 14:37 .ssh
-rw-r--r--  1 mininet mininet   10 Feb 10  2021 .sudo_as_admin_successful
-rw-rw-r--  1 mininet mininet  223 Feb 10  2021 .wget-hsts
drwxrwxr-x  3 mininet mininet 4096 Feb 10  2021 .wireshark
drwxrwxr-x  3 mininet mininet 4096 Sep 26 05:01 work
-rw-r----- 1 mininet mininet 112 Sep 26 05:14 .Xauthority
mininet@mininet-vm:~$
```

Figure 18: Визуализация результатов эксперимента

В результате выполнения данной лабораторной работы я познакомился с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получил навыки проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

1. Mininet [Электронный ресурс]. Mininet Project Contributors. URL: <http://mininet.org/> (дата обращения: 17.11.2024).
2. IPerff [Электронный ресурс]. URL: <https://iperf.fr/>.